

Solenoidal polarization in dielectrics

In a homogeneous isotropic linear dielectric with relative dielectric constant κ , if the free charge density is zero, then:

- ① the polarization vector is solenoidal
- 2) the polarization charge density is constant
- 3) the polarization vector is uniform
- 4) The electric field is uniform

Definition of dielectric strength

Dielectric strength indicates:

- a. the max value of the density of charge polarization that can be present inside a dielectric (beyond which a discharge occurs inside the dielectric)
- ☒ b. the max value of the electric field that may be present inside a dielectric (beyond which a discharge occurs inside the dielectric)
- c. the max value of the potential difference that can be applied to the ends of a dielectric (beyond which a discharge occurs inside the dielectric)
- d. the max value of the polarization that can be present inside a dielectric (beyond which a discharge occurs inside the dielectric)

Dependencies of capacitance

The capacitance of a capacitor does not depend:

- a. from the value of the dielectric constant of the medium present between the capacitor plate
- b. from the shape of the plates
- c. from the distance between the plates
- d. from the charge present on the plates

Charges inside a conductor in equilibrium

Question: A conductor in electrostatic equilibrium can contain charges in its interior:

- ☒ a. no, because inside the field must be null
- b. yes, it depends on its shape
- c. it depends on the conductor
- d. depends on whether the conductor has cavities

Selected Option: (a) no, because inside the field must be null

Electric field near negative conductor surface

At electrostatic equilibrium, the $\mathbf{E}$ field near the surface of a charged conductor with a uniform negative surface charge density σ is:

- ☒ (a) Uniform, perpendicular to the surface, of the direction entering the conductor
- ☐ (b) Uniform, parallel to the surface, of the direction entering the conductor
- ☐ (c) Uniform, perpendicular to the surface, facing out from the conductor
- ☐ (d) Anywhere null

Properties of the induction vector $\mathbf{D}$

Which of the following properties of the induction vector $\mathbf{D}$ is correct?

- a. $\mathbf{D} = \mathbf{E} - \mathbf{P}$
- b. $\mathbf{D} = \epsilon_r \mathbf{E}$ for each type of dielectric material
- ☒ c. $\iint \mathbf{D} \cdot \mathbf{n} \, d\Sigma = Q$ where Q is the free charge inside the closed surface Σ with normal unit vector $\mathbf{n}$
- d. $\iint \mathbf{D} \cdot \mathbf{n} \, d\Sigma = 0$ where Σ is a closed surface with normal unit vector $\mathbf{n}$

Gauss's Law for Displacement Vector

The electric displacement vector $\mathbf{D}$, defined as $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$, is such that

a. $\nabla \cdot \mathbf{D} = 0$

b. $\nabla \cdot \mathbf{D} = \rho_f$ where ρ_f is the free charge density

c. $\nabla \cdot \mathbf{D} = \rho_f / \epsilon_0$ where ρ_f is the free charge density

d. $\nabla \cdot \mathbf{D} = \rho_p$ where ρ_p is the bulk polarization charge density

Sum of surface polarization charges

In a dielectric material the sum of the surface polarization charges is zero when

- ☒ a. the material is homogeneous
- b. Never
- c. the applied electric field has a particular value
- d. always

Electric field of two parallel charged planes

Question: Two indefinite parallel planes, distant d , possess a uniformly distributed electric charge, equal in magnitude and sign. Which of the following statements is true:

- a. The electrostatic field is maximum in the space between the two planes.
- b. The electrostatic field is zero in the space between the two planes.**
- c. The electrostatic field between the two planes has an intensity proportional to d .
- d. The electrostatic field is non-zero and uniform in the space between the two planes.

Selected Option: (b) The electrostatic field is zero in the space between the two planes.

Boundary conditions for dielectrics

Upon crossing the separation surface between two homogeneous and isotropic dielectrics, having different dielectric constant:

- a. the tangential component of the electric field and the normal component of the dielectric induction vector are conserved
- b. the normal component of the electric field and the tangential component of the dielectric induction vector are conserved
- c. the tangential component of both the electric field and the dielectric induction vector is conserved
- d. the normal component of both the electric field and the dielectric induction vector is conserved

Dielectric in an isolated capacitor

Question: A dielectric is inserted inside a plain capacitor. Which statement is true, with the same charge distributed on the slabs:

- a. The capacity of the capacitor remains constant.
- b. The electrostatic field inside the dielectric is zero.
- c. The electrostatic field inside the capacitor remains constant.
- d. The dielectric induction field D inside the capacitor remains constant.**

Selected Option: (d) The dielectric induction field D inside the capacitor remains constant.

Energy stored in a capacitor

Calculate the energy stored in a capacitor of $10\text{ }\mu\text{F}$ capacity knowing that the charge present on the armatures is $Q = 1\text{ }\mu\text{C}$.

a. $U = 0.1\text{ J}$

b. $U = 0.5\text{ J}$

c. $U = 10^{-2}\text{ J}$

d. $U = 5 \cdot 10^{-8}\text{ J}$

Electric field outside a uniformly charged sphere

Given a sphere with a uniform charge density ρ , $\mathbf{E}$ is the field outside the sphere:

- a. $\mathbf{E}$ is always null
- b. $\mathbf{E}$ grows linearly as the distance r from the center increases
- c. $\mathbf{E}$ is always constant
- d. $\mathbf{E}$ is equivalent to that generated by a point charge placed in the center

Energy stored in a parallel plate capacitor

At the ends of a capacitor with flat and parallel plates, with an area $S = 0.02 \text{ m}^2$ and distant $d = 1 \text{ mm}$, there is a potential difference $V = 1000 \text{ V}$. In the hypothesis of operating in vacuum, the electrostatic energy stored in the capacitor is valid:

- a. $U_e = 8.85 \cdot 10^{-8} \text{ J}$
- ☒ b. $U_e = 8.85 \cdot 10^{-5} \text{ J}$
- c. $U_e = 1.77 \cdot 10^{-4} \text{ J}$
- d. $U_e = 1.77 \cdot 10^{-7} \text{ J}$

Polarization in a dielectric capacitor

A parallel plane plate capacitor is entirely filled with an homogeneous and isotropic dielectric material with relative dielectric constant $\kappa_r = 3$ placed perpendicularly to an external electric field of module $E_0 = 6 \text{ V/m}$. How much is worth the polarization module P ?

- a. $P = 3.54 \cdot 10^{-11} \text{ C/m}^2$
- b. $P = 3.54 \cdot 10^{-10} \text{ C/m}^2$
- c. $P = 5.31 \cdot 10^{-11} \text{ C/m}^2$
- d. $P = 10.62 \cdot 10^{-11} \text{ C/m}^2$
- e. $P = 8.85 \cdot 10^{-11} \text{ C/m}^2$

Capacitor with dielectric (constant charge)

The gap between the plates of a capacitor, initially empty, is completely filled with a dielectric material. With the same charge on the plates:

- (a) The capacitance decreases, the electrostatic field increases
- (b) The capacitance and the electrostatic field decrease
- (c) The capacitance and the electrostatic field increase
- ☒ (d) The capacitance increases, the electrostatic field decreases

Inserting a metal plate in a capacitor

A flat capacitor is charged through a d.d.p. (potential difference) supplied by a generator always connected with the capacitor. A metal plate of finite thickness is inserted between the slabs. Which of the following statements is true:

- 1) 1) The capacity of the capacitor remains constant.
- 2) 2) The charge on the capacitor plates remains constant.
- 3) 3) The capacitance of the capacitor decreases.
- 4) 4) The capacitance of the capacitor increases.

Saturation of electrical polarization

Considering the phenomenon of electrical polarization in a generic dielectric material, which of the following statements is correct.

- a. The polarization vector $\mathbf{P}$ is never parallel to the electric field vector $\mathbf{E}$ within the material.
- b. The modulus of the electric induction vector $\mathbf{D}$ represents a polarization charge per unit area.
- c. The polarization vector $\mathbf{P}$ is always parallel to the electric field vector $\mathbf{E}$ within the material.
- ☒ d. Above a threshold value of the electric field E , the modulus of the polarization vector $\mathbf{P}$ tends asymptotically to a maximum saturation value.

Flux of dielectric induction vector $\mathbf{D}$ through a surface

A point charge $q = 10^{-7}$ C is placed at the center of a sphere of dielectric material (homogeneous and isotropic) with a radius $R = 0.1$ m and having a relative dielectric constant equal to 1.5. What is the flux of the dielectric induction vector $\mathbf{D}$ through the dielectric surface?

a. $\Phi(\mathbf{D}) = 1.13 \cdot 10^4$ V m

b. $\Phi(\mathbf{D}) = 7.53 \cdot 10^3$ V m

☒ c. $\Phi(\mathbf{D}) = 10^{-7}$ C

d. $\Phi(\mathbf{D}) = 1.26 \cdot 10^{-8}$ C m²

Relation between polarization and displacement vectors

In a homogeneous isotropic linear dielectric with relative dielectric constant κ the polarization vector $\mathbf{P}$ is related to the vector $\mathbf{D}$ by the relation:

a. $\mathbf{P} = \frac{\kappa - 1}{\kappa} \mathbf{D}$

b. $\mathbf{P} = \frac{1}{\kappa} \mathbf{D}$

c. $\mathbf{P} = (\kappa - 1) \mathbf{D}$

d. $\mathbf{P} = \frac{\kappa}{\kappa - 1} \mathbf{D}$

Calculation of polarization charges

I have specified the polarization vector, relating to a given insulating material, so is it always possible to calculate the surface and volume polarization charges?

- a. Neither surface nor volume ones unless I also specify the external electric field
- b. Only the surface ones because I didn't specify the shape
- c. Just the surface charges but I have also to specify the external electric field
- ☒ d. Only the volume ones because I didn't specify the shape
- e. Yes always

Polarization Charge and Isotropy

A charged plane capacitor with plates parallel to the yz plane is filled with a dielectric material whose polarization vector is $\mathbf{P} = 2\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}$ (C/m²). How many are the charge densities of surface polarization σ_p and volume ρ_p ? If the charges on the capacitor plates are homogeneously distributed with constant charge density σ_0 , is the dielectric isotropic?

- a. $|\sigma_p| = 2$ C/m²; $\rho_p = 0$ C/m³; the dielectric is isotropic
- b. $|\sigma_p| = 2$ C/m²; $\rho_p = 6$ C/m³; the dielectric is not isotropic
- ☒ c. $|\sigma_p| = 2$ C/m²; $\rho_p = 0$ C/m³; the dielectric is not isotropic
- d. $|\sigma_p| = 2$ C/m²; $\rho_p = 6$ C/m³; the dielectric is isotropic
- e. $|\sigma_p| = 2$ C/m²; $\rho_p = 14$ C/m³; the dielectric is not isotropic

Polarization in a dielectric-filled capacitor

An empty capacitor with flat parallel faces of area Σ , distant d is connected to an electric generator of voltage V_0 . It is then introduced into the condenser a plate of relative dielectric constant κ of thickness d , so as to completely fill the capacitor. How much is the electrical polarization in the plate after insertion:

1. $\varepsilon_0(\kappa - 1) \frac{V_0}{d}$

2. $\varepsilon_0(\kappa - 1) \frac{V_0}{\kappa d}$

3. $\kappa \varepsilon_0 \frac{V_0}{d}$

4. $(1 + \kappa) \frac{V_0}{d}$

Energy of a dielectric-filled capacitor at constant voltage

An empty parallel plane capacitor with slabs of area Σ distant d , is connected to an electric generator of voltage V_0 . In this situation the electrostatic energy in the capacitor is $U_0 = C_0 V_0^2/2$, where $C_0 = \epsilon_0 \Sigma/d$. Subsequently a plate of relative dielectric constant κ and thickness d is introduced into the capacitor so as to fill the condenser completely. How much is the final electrostatic energy in the capacitor?

1. κU_0
2. U_0
3. $\frac{U_0}{\kappa}$
4. $(1 + \kappa)U_0$

Conservative nature of $\mathbf{D}$ vector

In a dielectric material the induction vector is $\mathbf{D} = \epsilon \mathbf{E}$. This means that

- (a) the integral of $\mathbf{D}$ along a closed path is always zero
- (b) $\mathbf{D}$ is conservative
- (c) the flow of $\mathbf{D}$ through a generic surface is zero
- ☒ (d) $\mathbf{D}$ may or may not be conservative

Calculation of polarization charges

I have specified the polarization vector, relating to a given insulating material, so is it always possible to calculate the surface and volume polarization charges?

- a. Neither surface nor volume ones unless I also specify the external electric field
- b. Only the surface ones because I didn't specify the shape
- c. Just the surface charges but I have also to specify the external electric field
- ☒ d. Only the volume ones because I didn't specify the shape
- e. Yes always

Polarization surface charge densities in a dielectric sheet

In a sheet of dielectric material thickness d , with the limiting surfaces parallel to plane y, z located at $x = 0$ and $x = d$, the polarization vector is $\mathbf{P} = \alpha x \mathbf{u}_x + \beta \mathbf{u}_y$, where α and β are constant. What are the polarization surface charge densities at $x = 0$ and at $x = d$?

1. 0 and αd
 2. α and β
 3. αd and $-\alpha d$
 4. $\alpha/2$
-

Polarization Charge and Isotropy

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- ☒ c. $|\sigma_p| = 2$ C/m²; $\rho_p = 0$ C/m³; the dielectric is not isotropic
- d. $|\sigma_p| = 2$ C/m²; $\rho_p = 6$ C/m³; the dielectric is isotropic
- e. $|\sigma_p| = 2$ C/m²; $\rho_p = 14$ C/m³; the dielectric is not isotropic

Volume polarization charge density

In an anisotropic dielectric material the polarization vector is $\mathbf{P}(x, y, z) = 2ax\mathbf{i} - 3bz\mathbf{j} + cy\mathbf{k}$ (where a, b , and c are constants suitable size). Determine the value of the volume polarization charge density ρ_P present in the dielectric material.

a. $\rho_P = 3b$

b. $\rho_P = -3b\epsilon_0$

c. $\rho_P = (2a - 3b + c)\epsilon_0$

d. $\rho_P = -2a$

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d. $\mathbf{P} = \frac{\kappa}{\kappa - 1} \mathbf{D}$

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Question: An empty parallel plane capacitor with slabs of area Σ distant d , is connected to an electric generator of voltage V_0 . In this situation the electrostatic energy in the capacitor is $U_0 = C_0 V_0^2 / 2$, where $C_0 = \epsilon_0 \Sigma / d$. Subsequently a plate of relative dielectric constant κ and thickness d is introduced into the capacitor so as to fill the condenser completely. How much is the final electrostatic energy in the capacitor?

- a. κU_0
- b. U_0
- c. $\frac{U_0}{\kappa}$
- d. $(1 + \kappa)U_0$

Selected Option: (a) κU_0

Conservative nature of $\mathbf{D}$ vector

In a dielectric material the induction vector is $\mathbf{D} = \epsilon \mathbf{E}$. This means that

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- (b) $\mathbf{D}$ is conservative
- (c) the flow of $\mathbf{D}$ through a generic surface is zero
- ☒ (d) $\mathbf{D}$ may or may not be conservative